Chenxu Niu

AI-native Solutions Architect and HPC System Engineer, NVIDIA
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Research Interests

High-performance computing systems, energy-efficient computing, AI inference on HPC systems, and scientific data management.

Education

Ph.D. in Computer Science, Texas Tech University Jun 2019 - Dec 2025

Lubbock, Texas

Dissertation: Accelerating Scientific Dataset Discovery with Semantic Metadata Search and Large Language Models.

M.S. in Information Science and Technology, University of Science and Technology of China Sep 2015 - Jun 2018

Hefei, China

B.E. in Mathematics, University of Science and Technology of China Sep 2011 - Jun 2015

Hefei, China

Professional Experience

AI-native Solutions Architect and HPC System Engineer Jan 2026 - Present

NVIDIA

- Work with AI-native, consumer internet, and enterprise customers on GPU server, networking, and data center deployments.
- Guide network, compute, and storage design discussions and support server and cluster bring-up.
- Analyze and debug compute and network configuration and performance issues.

HPC Systems Administrator and Research Assistant Jun 2019 - Dec 2025

Texas Tech University · Lubbock, Texas

- Administered Slurm and the software stack for research clusters serving hundreds of users.
- Built monitoring infrastructure for power, thermal, and resource utilization using iDRAC, Redfish, and Grafana.
- Automated cluster deployment and updates with Ansible; deployed distributed LLM training and inference on GPU clusters.

Peer-Reviewed and Accepted Publications

2026

[10] H. Chen, X. Liu, Y. Liu, J. Jiang, X. Liu, **C. Niu**, and B. He. **The 1/W Law: Context Length is the Dominant Energy Lever in LLM Inference Fleets**. *Conference on Neural Information Processing Systems (NeurIPS '26)*, 2026. Accepted; proceedings version forthcoming. PDF · BibTeX

[9] J. Xiong, R. Li, **C. Niu**, G. Gao, C. Xing, Y. Zhang, X. Wan, J. Cui, C. Bai, Y. Hua, Y. Wang, M. Ling, X. Wang, and T. Xie. **MicroEvo: Knowledge-Guided LLM Sampling for Efficient Microarchitecture Design Space Exploration**. *IEEE/ACM International Conference on Computer-Aided Design (ICCAD '26)*, 2026. Accepted; arXiv v1, proceedings version forthcoming. PDF (arXiv v1) · arXiv · Code · BibTeX

[8] Y. Zhao, J. Li, **C. Niu**, A. Sill, and Y. Chen. **Power-Centric Observability for HPC Systems: Design, Deployment, and Evaluation on REPACSS**. *Proceedings of the Practice and Experience in Advanced Research Computing 2026 (PEARC '26)*, pp. 1-8, 2026. PDF · DOI · BibTeX

[7] **C. Niu**, W. Zhang, J. Li, Y. Zhao, T. Wang, X. Wang, and Y. Chen. **TokenPowerBench: Benchmarking the Power Consumption of LLM Inference**. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI '26), 40(38)*, pp. 32582-32590, 2026. Acceptance rate: 17.6%. PDF · Code · DOI · BibTeX · Poster

[6] G.-W. Wan, S. Wong, S. Su, **C. Niu**, N. Wang, X. Wan, Q. Chen, M. Xing, J. Zhang, J. Ye, Y. Wang, R. Song, T. Ni, Q. Xu, N. Guan, Z. Jiang, X. Wang, Y. Chen, and J. Yang. **FIXME: Towards End-to-End Benchmarking of LLM-Aided Design Verification**. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI '26), 40(2)*, pp. 1087-1095, 2026. Acceptance rate: 17.6%. PDF · DOI · BibTeX

2025

[5] **C. Niu**, W. Zhang, Y. Zhao, and Y. Chen. **Energy Efficient or Exhaustive? Benchmarking Power Consumption of LLM Inference Engines**. *ACM SIGEnergy Energy Informatics Review, 5(2)*, pp. 56-62, 2025. PDF · Code · DOI · BibTeX

[4] **C. Niu**, W. Zhang, M. Side, and Y. Chen. **ICEAGE: Intelligent Contextual Exploration and Answer Generation Engine for Scientific Data Discovery**. *Proceedings of the 37th International Conference on Scalable Scientific Data Management (SSDBM '25)*, pp. 1-10, 2025. PDF · DOI · BibTeX

2023

[3] **C. Niu**, W. Zhang, S. Byna, and Y. Chen. **PSQS: Parallel Semantic Querying Service for Self-Describing File Formats**. *IEEE International Conference on Big Data (BigData '23)*, pp. 536-541, 2023. PDF · DOI · BibTeX

2022

[2] **C. Niu**, W. Zhang, S. Byna, and Y. Chen. **Kv2vec: A Distributed Representation Method for Key-Value Pairs from Metadata Attributes**. *IEEE High Performance Extreme Computing Conference (HPEC '22)*, pp. 1-7, 2022. PDF · DOI · BibTeX

2019

[1] W. Zhang, S. Byna, **C. Niu**, and Y. Chen. **Exploring Metadata Search Essentials for Scientific Data Management**. *IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC '19)*, pp. 83-92, 2019. PDF · DOI · BibTeX

Preprints

- **C. Niu**. **TokenPowerSandbox: Evidence-Gated CPU-First Screening for Energy-Aware LLM Serving**. *arXiv preprint 2608.18149 (v1)*, 2026. PDF (arXiv v1) · arXiv · DOI · BibTeX

Research Projects

HPC Data Center Design and Implementation, NSF REPACSS Sep 2023 - Dec 2025

- Led design and deployment work for a renewable energy-powered HPC facility.
- Implemented an iDRAC, Redfish, Grafana, and TimescaleDB monitoring stack to track power, carbon, thermal, and utilization metrics across more than 1,000 sensors.
- Worked on 200 Gbps InfiniBand and 100 Gbps Ethernet network topology for AI and scientific workloads.

Distributed LLM Inference Monitoring and Optimization Sep 2023 - Jun 2025

- Built monitoring dashboards for GPU utilization, memory bandwidth, power draw, and thermal throttling.
- Benchmarked vLLM, TensorRT-LLM, and Ray Serve across hardware configurations and model sizes.
- Developed TokenPowerBench to measure the power consumption and performance of open-source LLM inference.

Professional Service

- **Program Committee:** AAAI 2027, AAAI 2026, BigData 2026, PEARC 2026.
- **Reproducibility Committee:** SC 2025.
- **Paper Reviewer:** NeurIPS Workshop 2026, ACM TiiS 2026, BigData 2025, CCGrid 2024, SSDBM 2024.
- **Conference Volunteer:** SC 2021 and SC 2024.

Teaching Experience

Graduate Teaching Assistant, Texas Tech University 2020 - 2022

- Computational Thinking with Data Science (Fall 2021 - Fall 2022).
- Advanced Operating System Design (Spring 2021).
- Analysis of Algorithms (Spring 2020).

Invited Talks and Presentations

Semantic Search and Natural Language Query over HDF5. HDF5 User Group Meeting (HUG24), 2024.

Technical Skills

Programming and DevOps: Python, C/C++, CUDA, Docker, Kubernetes, Terraform, Ansible, Git.

HPC and infrastructure: Slurm, Warewulf, Spack, InfiniBand, Lustre/GPFS, MPI, OpenMP, Linux administration.

AI and monitoring: PyTorch, vLLM, TensorRT-LLM, Ray, DeepSpeed, Prometheus, Grafana, Redfish.